

LESSON 6-5 • SESSION 1 • PRACTICE • FORM A

Writing Algebraic Expressions

Learning target – I can write an algebraic expression that uses a variable for an unknown amount. · 6.EE.A.2a

REMEMBER

- A **variable** is a letter for an unknown amount. $5n$ means $5 \times n$.
- In $5n + 12$, 5 is the **coefficient** and 12 is the **constant**.
- An amount that **repeats** goes with the variable. “4 less than x ” is $x - 4$.

WORKED EXAMPLE

A taxi charges \$3 to start plus \$2 for each mile. Write an expression for the fare for m miles.

Step 1 Let m = the number of miles.

Step 2 \$2 repeats each mile: $2m$.

Step 3 The \$3 start fee happens once. Join: $2m + 3$.

Answer $2m + 3$ dollars

WORDS **variable** (*variable*) · **coefficient** (*coeficiente*) · **constant** (*constante*) · **algebraic expression** (*expresión algebraica*)

PRACTICE

show your work · label units

1. Write an algebraic expression for each phrase.

a. 7 more than a number n _____

b. 5 times a number x _____

c. 4 less than y _____

d. w divided by 3 _____

e. twice t , plus 1 _____

f. 6 less than the product of 3 and k _____

2. **Movie Club.** A movie club buys one \$6 popcorn for each member. It also pays one \$12 delivery charge. Let m stand for the number of members.

a. Write the popcorn cost as a term. Name its coefficient. _____

b. What is the constant? Why has it no variable? _____

c. Write one expression for the total cost. _____

3. **Concert Tickets.** Each ticket costs \$8. There is one \$25 service fee for the whole order. Write an expression for the cost of t tickets.

Hint The \$8 repeats; the \$25 happens once.

Show your work.

Expression _____

4. **Car Wash.** A team charges \$7 for each car it washes. It pays one \$45 fee for supplies. Write an expression for the money left after c cars.

Hint Money collected minus the fee.

Show your work.

Expression _____

5. **Gym Membership.** A gym charges \$17 each month plus a one-time \$40 joining fee. Let m be the number of months. Write an expression for the cost.

Hint The monthly charge goes with the variable.

Show your work.

Expression _____

6. **Daily Rental.** A kayak rental costs \$5 for each day. Rani wrote $5 + d$ for the cost of d days. Justify or correct her expression.

Hint Does “for each day” mean add or multiply?

Show your work.

Expression _____

7. **Two Items.** A stand sells lemonade for \$3 a cup and cookies for \$2 each. Let c be the cups sold and k be the cookies sold. Write one expression for the money collected.

Hint Write a term for each item, then add.

Show your work.

Expression _____

8. **Compare Gyms.** Gym A charges a \$40 joining fee plus \$25 each month.
- Write an expression for Gym A’s cost after m months. _____
 - Gym B charges no joining fee but \$30 each month. Write its expression. _____
 - Which gym costs less for 6 months? Show how you know. _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Bela’s Work.** Books cost \$11 each and shipping is \$6 one time. Bela wrote $11 + 6b$ for b books. Explain her mistake. Then write the correct expression.

TEST PRACTICE

choose the best answer

10. “7 more than a number n ” is written –
- A** $n + 7$ **B** $7n$ **C** $n - 7$ **D** $7 - n$
11. In the expression $6p + 9$, the variable is –
- A** 6 **B** p **C** 9 **D** 15

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

40m + 17 X	8t + 25 Y	25m + 40 L	45 – 7c Z	6m + 12 S	7c – 45 M	5 + d Q	17m + 40 B	5d O
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#2c	#3	#4	#5	#6	#8a
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CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____